Gerald Nwogbo Okeke

AI Engineer | Solutions Architect

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Abuja, Nigeria

ABOUT ME

A results-driven AI Engineer and Solutions Architect with 12+ years of experience designing and delivering production grade AI platforms, cloud infrastructure, and agentic AI systems. Passionate about building scalable, responsible AI ecosystems that empower both engineers and non-technical stakeholders to create transformative technology.

PROGRAMMING

Python (FastAPI, Django, Flask)

TypeScript / JavaScript

NodeJS, NestJS, Fastify

C#, VB.Net / ASP.Net Core

SQL · Bash Script

AI & ML

LangChain / LangGraph

RAG Pipeline Architecture

LLM Orchestration (GPT, Claude, Gemini, Grok)

Prompt Engineering & Versioning

Agentic AI / Tool-use Systems

Vector Databases (Qdrant, Chroma)

CLOUD & DEVOPS

Microsoft Azure (ACA, Functions, PostgreSQL, CosmosDB)

AWS (Lambda, EKS, RDS, DynamoDB, CloudWatch, Cloudfront)

PROFESSIONAL SUMMARY

Seasoned AI Engineer and Solutions Architect with over 12 years of experience delivering production-grade cloud-based AI systems across regulated sectors including government, nuclear, and financial services. Deep hands-on expertise in agentic AI development, LLM orchestration, RAG pipelines, and scalable multi-tenant platform design. Proven ability to translate complex product visions into reliable AI ecosystems — from data infrastructure and model pipelines to AI-assisted development workflows. Adept at enabling non-programmers to participate in AI-driven software creation through agent-builder platforms and AI-first tooling.

WORK EXPERIENCE

Software Engineer | DevOps Engineer | Nigerian Nuclear Regulatory Authority (NNRA) / *Nov 2013 – Present*

Winnipeg, Manitoba, Canada (Remote) | Abuja, Nigeria

- Architected and deployed AI-powered regulatory information systems serving the national nuclear authority, integrating multi-model AI pipelines (GPT, Claude) with FastAPI backends and TypeScript-based frontends to deliver intelligent document processing and regulatory compliance automation.
- Designed and implemented agentic AI workflows that automate internal processes, enabling non-engineering staff to configure and trigger intelligent automation routines without writing code directly enabling a no-code AI-builder environment across the authority.
- Built and maintained RAG-based knowledge retrieval systems, integrating document stores with vector search capabilities to surface relevant regulatory data in real time across internal platforms.
- Led the design and operation of scalable cloud infrastructure on AWS including Lambda serverless functions, EC2 autoscaling groups, EKS Kubernetes clusters, CloudFront CDN, and Route 53 applying infrastructure-as-code via CloudFormation for reproducible provisioning.
- Implemented end-to-end CI/CD pipelines using GitHub, CircleCI, and Travis CI, reducing release cycles and establishing automated regression testing and deployment validation
- Designed and managed containerised microservices using Docker and Kubernetes (EKS), breaking down monolithic applications to improve resilience, scalability, and developer velocity.
- Established operational standards including SLIs, SLOs, monitoring dashboards (AWS CloudWatch and X-Ray), alerting thresholds, and incident runbooks to maintain production reliability.
- Enforced security-by-design principles including IAM roles with least-privilege access, encryption at rest and in transit, secrets management, and strict alignment with the NNRA ICT Security Policy informed by PHI/PII data handling best practices.
- Configured and managed relational database infrastructure across MSSQL and PostgreSQL, authored complex stored procedures for regulatory reporting, and implemented the 3-2-1 backup model for data continuity.
- Provided executive-level technical advisory to the DG/CEO on emerging AI and ICT trends, influencing strategic adoption of AI-assisted tooling and cloud-native technologies across the authority.

Data Engineer & Software Developer | FinTrak Software / *Feb 2012 – Nov 2013*

Lagos, Nigeria

- Led the data engineering and software development workstream for an IFRS-compliant financial reporting platform deployed across multiple commercial banks in Nigeria, a high-stakes, regulated production environment requiring zero-error financial data transformation.
- Designed and built robust ETL pipelines using Microsoft SSIS to extract, transform, and load financial data from Oracle core banking systems into MSSQL data warehouses, forming the data infrastructure backbone for AI-ready reporting.
- Developed complex Oracle and MSSQL data migration scripts to reconcile and move financial records across banking systems, ensuring data lineage, accuracy, and auditability foundational to responsible data governance.
- Authored stored procedures to calculate amortised costs and loan impairment values per IFRS standards, delivering financial computation logic that directly powered executive-level reporting and compliance dashboards.
- Built the IFRS conversion application in VB.NET, enabling Nigerian banks to transition from legacy SAS-based to IFRS-compliant reporting systems demonstrating full-stack delivery capability across data and application layers.
- Delivered client onboarding, training, and post-deployment early-life support with strict SLA adherence, reinforcing a track record of reliable production system ownership.

Junior Data Engineer & Software Developer | DABAROBJECTS SOLUTION / *Jun 2010 – Jan 2012*

Abuja, Federal Capital Territory, Nigeria

- Designed the MySQL database architecture for MADEX (Mobile Electronic Data Exchange), a national mobile data collection platform for the Federal Government agency responsible for primary healthcare infrastructure in Nigeria.

Terraform / CloudFormation
Docker · Kubernetes (EKS)
CI/CD: GitHub Actions, CircleCI, Travis CI, Jenkins

DATA & DATABASES

PostgreSQL · MSSQL · MySQL
MongoDB · CosmosDB · DynamoDB
Oracle · ETL Pipelines
Feature Stores · Data Lineage

ACHIEVEMENTS

🏆 5+ AI products shipped to production across regulated industries
🏆 Led end-to-end AI platform architecture for Nigeria's national nuclear regulatory body
🏆 Deployed IFRS compliance AI platform across commercial banks in Nigeria

EDUCATION

M.Sc. Computer Science
Nile University of Nigeria · 2019

B.Tech. Information Mgt Technology
FUTO · 2010

Cloud DevOps Engineer
Udacity · 2022

PCEP – Python Programmer
Python Institute · Certified

CERTIFICATIONS

• AI Engineer Agentic Track: The Complete Agent & MCP Course
• AI Leader: Generative AI & Agentic AI for Leaders & Founders
• AI Engineer Core Track: LLM Engineering, RAG, QLoRA, Agents
• PCEP Python Entry-Level Programmer
• Cloud Developer (Udacity)
• Cloud DevOps Engineer (Udacity)
• Oracle DBA – SunSolaris OS
• What Is Generative AI?
• Verified Intl Academic Qualifications

- Built and documented database schemas to support field health data capture and structured reporting, ensuring data quality and traceability in a government production environment.
- Conducted comprehensive software testing, identifying bugs and security vulnerabilities across the platform, and produced technical documentation to support operational handover and client support.
- Researched and proposed emerging technologies to enhance system capabilities, demonstrating early initiative in applied technology adoption.

AI PLATFORM & ARCHITECTURE HIGHLIGHTS

- Production Agentic AI: Designed and operated agent systems that enable non-programmers to build and configure AI-driven workflows without engineering involvement — mirroring agent-builder platform requirements.
- LLM Orchestration: Hands-on experience with LangChain and LangGraph for building LLM orchestration pipelines, prompt versioning systems, and multi-model experimentation frameworks (GPT, Claude, Gemini, Grok).
- RAG & Vector Search: Designed Retrieval-Augmented Generation pipelines integrating document indexing, vector database retrieval (Qdrant-compatible), and structured prompt engineering for domain-specific AI applications.
- AI-Assisted SDLC: Implemented AI-assisted coding workflows, automated test generation, and release validation automation — reducing manual development overhead and improving engineering velocity across teams.
- Azure Cloud Infrastructure: Practical expertise with Azure-managed services including containerised workloads, serverless functions, PostgreSQL managed databases, CosmosDB, and identity/secrets management.
- Data Governance & Privacy: Designed systems handling sensitive government and financial data with privacy-by-design principles, PII handling policies, data retention, and cross-border compliance considerations.
- Distributed Systems & API Design: Architected event-driven, microservice-based distributed systems with production-grade API design, error budgets, observability, and incident response capabilities.